YIMIAN LIU

Software Engineer | Agentic AI & Machine Learning
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EXPERIENCE

Guzman Energy Denver, CO

Operations Research Engineer (AI & Platform Tech Lead) Jan 2026 – Present

- Led a three-person AI-native team build Guzman's core internal trading, AI, and data platform from scratch in under 2 years: 32 production apps, 71% of employees active weekly, $2M daily trading volume, and 3M+ system events/day.
- Built an agentic auto-research system for load/generation forecasting; agents generated hypotheses, ran reproducible backtests, shared experiments/evidence across agents to guide new research, published immutable model artifacts; LLM judge filtered leaky runs and promoted candidates by realized performance, reducing MAE by 17% vs best vendor.
- Built and owned a fully-automated production SPP trading system that integrates forecasts with trader-in-the-loop formula-based strategy design; engineered reliability with preflight checks, state-drift detection, cross-region retries, and agentic anomaly detection, achieving 100% on-time submissions for 6+ months while submitting $1.1M+ daily.
- Proposed and led the adoption of Claude as the company's self-serve analysis and automation layer, growing weekly active usage from 0-82% of employees; led delivery of a unified MCP connector exposing internal/external data and APIs that make Claude really useful, with role-based access control and audit logging for data governance.
- Grew MCP events 2.8x MoM for four months by using AI agents to identify user pain points from audit logs, rapidly resolving user blockers, leading cross-department onboarding, and hosting biweekly company-wide AI Office Hours.
- Initiated and led delivery of multiple AI agentic systems, notably: a security scanner that found 6 critical vulnerabilities, including one in a third-party vendor system; a data anomaly and human-error detection system that helped traders proactively avoid $20K+ in losses; an RFP finder that helped Commercial surface a real opportunity in its first month.
- Empowered all three team members to achieve 6x traditional engineering productivity by building an AI-native, self-driven, research-to-production culture: encouraging end-to-end ownership, close end-user engagement, engineer-led innovation; hosting weekly AI research seminars; and automating code review, testing, and CI/CD.

Software Engineer Oct 2024 – Dec 2025

- Architected and built the platform foundation from scratch using Python/GCP, React/TypeScript, and BigQuery; team-owned data pipelines span 3.5B+ production rows; delivering core apps including trading submissions, AI data marketplace, position, settlement & risk management, resource planning map, ticket & notification system.

Elseware Capital | *Full Stack Developer (Remote)* Jan 2024 – Sep 2024

- Built from scratch a full-stack real estate trading Web3 app using React, FastAPI, and MySQL, supporting asset tokenization, KYC, portfolio, and live blockchain transactions; optimized APIs and database to reduce latency by 80%.

Ecesis Investment | *Frontend Developer Intern (Part-time, Remote)* Jul 2023 – Dec 2023

- Built a cross-platform AI scribing mobile app in React Native with real-time audio transcription and LLM summarization; optimized WebSocket streaming and Redux caching, reducing transcription latency 60% and UI rendering time 80%.

EDUCATION

Cornell University | M.Eng., Computer Engineering | GPA: 3.95/4.00 Aug 2022 – Dec 2023

University of Liverpool | B.Eng. (Honors), Electrical & Electronic Engineering | GPA: 3.97/4.00 Sep 2017 – Jul 2021

Coursework: Deep Learning (A+), Machine Learning (A), Computer Vision (A+), Distributed Computing Principles.

SELECTED PROJECTS

Transformer & Reinforcement Learning [Demo] Built a Transformer policy using supervised pretraining and PPO for predator–prey simulation; distilled the policy by 40% for <10 ms browser inference.

Distributed Systems [Code] Implemented a sharded Java key/value store using Multi-Paxos and two-phase commit for strongly consistent cross-shard transactions.

TECHNICAL SKILLS

AI & Machine Learning: Python, PyTorch, Time-Series Forecasting, Model/LLM Evaluation, RAG, MCP, SFT, RL, STT, SQL
Systems & Engineering: TypeScript, React, FastAPI, REST APIs, MySQL, OAuth, RBAC, GCP, AWS, K8s, Docker, CI/CD, Git